Comparative Metaheuristic and Reinforcement Learning Optimization for Cooperative Electric Vehicle Fleet Routing with Nonlinear Charging and Shared Station Capacity

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***Abstract*—This paper presents a comprehensive study on optimizing electric vehicle (EV) fleet routing and charging operations using four metaheuristic algorithms and a deep reinforcement learning approach. The problem is formulated as a multi-objective optimization task that minimizes both fleet makespan and total charging cost subject to nonlinear state-of-charge–dependent charging physics, shared station capacity with FIFO queuing, and discrete speed-level trade-offs. Simulated Annealing (SA), a Genetic Algorithm (GA), Particle Swarm Optimization (PSO) with adaptive inertia, Teaching-Learning Based Optimization (TLBO), and a Deep Q-Network (DQN) agent are implemented and evaluated across single-fork and double-fork network topologies with 3–7 heterogeneous EVs. PSO achieves the lowest weighted objective on uncongested scenarios (274.25 vs. 291.55 for GA), while GA demonstrates superior robustness under congested conditions (2321.90 vs. 3447.87 for PSO). TLBO provides a competitive parameter-free alternative, consistently outperforming SA by 10–25%. The DQN agent achieves the fastest makespans but incurs 183–199% higher charging costs than PSO. A hyperparameter grid search over 243 PSO configurations reveals that light inertia ($w = 0.3$–0.6) with symmetric cognitive/social coefficients yields optimal trade-offs, and adaptive inertia improves convergence by 16% over static schedules.**

***Index Terms*—Electric vehicles, fleet optimization, metaheuristics, particle swarm optimization, genetic algorithm, TLBO, reinforcement learning, nonlinear charging, multi-objective optimization.**

I. Introduction

The electrification of commercial vehicle fleets presents a complex operational challenge that extends well beyond traditional vehicle routing. Unlike conventional fleets, EV operations must account for battery state-of-charge (SOC) constraints, the nonlinear physics of lithium-ion charging, limited charging infrastructure, and the temporal coupling between vehicles competing for shared charging plugs [1].

This temporal coupling is the defining feature of the cooperative EV fleet routing problem: when multiple vehicles share a corridor with capacity-constrained stations, one vehicle's charging decision directly affects the waiting times and feasibility of all subsequent arrivals. Uncoordinated charging creates congestion at popular stations, leading to cascading delays and suboptimal fleet performance [2].

Recent literature has addressed various facets of this problem. Froger et al. [3] introduced a flow-based formulation for deterministic station capacity in the E-VRP-NL-C. Kancharla and Ramadurai [4] extended the model with load-dependent discharging rates. Keskin et al. [5] modeled stochastic waiting times using M/G/1 queueing theory, while Karakatič [6] proposed a two-layer genetic algorithm for multi-depot scenarios with nonlinear charging.

This work makes four contributions: (1) a unified problem formulation incorporating nonlinear SOC-dependent charging, FIFO queuing at capacity-constrained stations, and discrete speed levels; (2) implementation and comparative analysis of four metaheuristic algorithms—SA, GA, PSO with adaptive inertia, and TLBO; (3) development of a Deep Q-Network reinforcement learning approach as an alternative machine learning solution; and (4) extensive experimental evaluation across seven scenarios of increasing complexity on single-fork and double-fork network topologies.

II. Problem Formulation

A. Network Model and Decision Variables

The transportation network is modeled as a directed graph $G = (V, E)$ where vehicles travel from origin A to destination B through fork-shaped topologies offering route choices at each junction. Each edge $e = (u, v) \in E$ has associated travel

time $T_{uv}(s)$ and energy consumption $E_{uv}(s)$ as functions of the discrete speed level $s \in \{1, \ldots, 5\}$.

For each vehicle i in the fleet, the decision variables are: binary route branch selection r_i; binary charging indicator $x_{i,s} \in \{0, 1\}$ at each station s; continuous energy purchased $E_{i,s} \geq 0$ (kWh); and integer speed levels per edge. Table I summarizes the speed-energy trade-off.

TABLE I: Discrete Speed Level Options

Level	Description	Time Factor	Energy Factor
1	Very Slow	$+40\%$	-25%
2	Slow	$+20\%$	-10%
3	Normal	Baseline	Baseline
4	Fast	-15%	$+20\%$
5	Very Fast	-30%	$+50\%$

B. Objectives and Constraints

The weighted objective combines makespan minimization and cost minimization:

$$f = w_t \cdot \underbrace{\max_{i \in \mathcal{V}} t^{\text{arr}}_{i,B}}_{f_1\text{: makespan}} + w_c \cdot \underbrace{\sum_{i \in \mathcal{V}} \sum_{s \in \mathcal{S}} p_s E_{i,s}}_{f_2\text{: charging cost}} \tag{1}$$

where $w_t = w_c = 1.0$ in all experiments. The problem is subject to five constraint classes:

C1—SOC Bounds: $0.10 \leq \text{SOC}_i(t) \leq 1.00$ for all vehicles at all times.

C2—Energy Balance: $\text{SOC}_{i,v+1} = \text{SOC}_{i,v} + E_{i,v}/B_i - \varepsilon_{(v,v+1)}/B_i$, ensuring SOC consistency along each route.

C3—Station Capacity: $\sum_i z_{i,s}(t) \leq C_s$ for all stations s and times t, enforcing concurrent plug limits with FIFO queuing.

C4—Nonlinear Charging: Charging power follows $P(\text{SOC}) = \min(P_{\text{curve}}(\text{SOC}), P_{\text{station}}, P_{\text{vehicle}})$, with tapering above 80% SOC. Charging time is computed via numerical integration with efficiency $\eta = 0.95$.

C5—Route Connectivity: Each vehicle must follow valid edges from A to B.

III. Methodology

All algorithms share a common `VehiclePlan` representation encoding per-vehicle routes, charging amounts, and speed levels. A unified feasibility repair module enforces battery limits, plug capacities, and FIFO queuing before evaluation.

A. Simulated Annealing

SA is a trajectory-based metaheuristic [7] that accepts worse solutions with probability $P = \exp(-\Delta f/T)$, where temperature T decreases geometrically: $T_{k+1} = \alpha \cdot T_k$ with $\alpha = 0.995$. Neighborhood operators include speed adjustment ($\pm$1–2 levels), charge tweaking, route toggling, and station addition/removal. Infeasible candidates trigger targeted repairs based on specific violation codes.

B. Genetic Algorithm

The GA [8] maintains a population of candidate fleet plans with tournament selection (size 3), two-point crossover, and mutation (rate 0.1) including route branch flips and charging perturbations. Elitism preserves the top 10–15% of individuals. Infeasible offspring receive a heavy penalty (fitness $\approx -10{,}000$) and undergo targeted repair. Diversity tracking monitors population standard deviation and stagnation count per generation.

C. Particle Swarm Optimization with Adaptive Inertia

PSO [9] updates particle velocities and positions according to:

$$v^{t+1} = w_t v^t + c_1 r_1 (p^* - x^t) + c_2 r_2 (g^* - x^t) \tag{2}$$

where p^* and g^* are personal and global bests, and $r_1, r_2 \sim U(0, 1)$. The key innovation is an *adaptive inertia schedule*: w linearly decays from 0.9 to 0.4, with exploration spikes triggered when the global best stagnates. Continuous particle positions are mapped to discrete route/charging choices via the shared repair module. Fig. 1 demonstrates that adaptive inertia reduces the weighted objective by 16% compared to a static schedule (1483.6 vs. 1760.8 after 150 iterations).

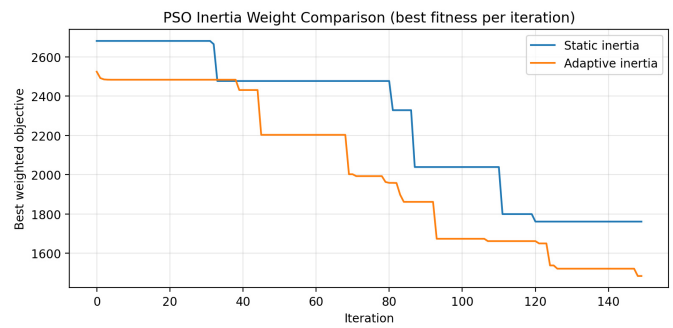

Fig. 1: Static vs. adaptive inertia convergence on the double-fork scenario. Adaptive inertia achieves 16% lower final objective.

D. Teaching-Learning Based Optimization

TLBO, proposed by Rao et al. [10], requires *no algorithm-specific parameters*. A hybrid blending operator handles mixed variables: for continuous decisions (charging, speed), $x_{\text{new}} = x_{\text{old}} + \alpha \cdot (x_{\text{target}} - x_{\text{old}})$; for discrete variables (routes), probabilistic selection is applied based on the blending factor α. In the Teacher Phase, the best solution guides all learners; in the Learner Phase, individuals improve through peer interaction. Greedy selection in both phases ensures monotonic fitness improvement.

E. Deep Q-Network Reinforcement Learning

The DQN approach formulates EV routing as a Markov Decision Process with a 25-dimensional state space (node encoding, SOC levels, station occupancy) and compound actions (fork choices $\times$ SOC targets $\times$ speed levels). A Dueling DQN architecture [11] with Prioritized Experience Replay [12] and

Double DQN updates [13] is trained for 500 episodes. Guided exploration combines expert-biased actions (probability 0.7ϵ) with uniform random exploration (0.3ϵ).

IV. Experimental Setup

A. Network Topologies

Two topologies are evaluated: (1) a *single-fork corridor* (A→J→{Upper: S1→S2, Lower: S3}→M→B) and (2) a *double-fork network* with two consecutive fork-merge segments offering expanded routing choices.

B. Case Studies

Seven scenarios of increasing complexity are tested, summarized in Table II.

TABLE II: Case Study Configurations

Scenario	EVs	Stations	Pricing
Single-Small	3	S1/S2/S3	Standard
Single-Large	7	S1/S2/S3	Standard
Double-Sparse	4	6 stations	Standard
Double-Balanced	5	6 stations	Standard
Double-Premium	4	6 stations	Standard
Double-Congested	6	-1 plug/stn	Standard
Double-High Demand	7	S2/S5 upgraded	$+15\%$

Algorithm parameters are set as: SA ($T_0 = 1000$, $\alpha = 0.995$, 2000 iterations); GA (pop=50, 100 generations, $p_m = 0.1$); PSO (swarm=50, 100 iterations, w: 0.9→0.4); TLBO (pop=60, 150 iterations, *no other parameters*); DQN (500 episodes, $\gamma = 0.99$, ϵ: 1.0→0.05).

V. Results and Discussion

A. Metaheuristic Performance Comparison

Table III presents results for SA, GA, PSO, and TLBO across five network configurations. PSO achieves the best weighted objective in 4 of 5 scenarios, with improvements of 6–40% over the next-best algorithm. GA shows the fastest makespans in larger fleet scenarios, while TLBO consistently outperforms SA by 10–25% despite having no tunable parameters.

TABLE III: Metaheuristic Performance (Selected Scenarios)

Scenario	Alg.	f	Mksp (min)	Cost (EGP)
Single-Small	SA	757.2	304.2	453.1
	GA	291.6	271.6	20.0
	PSO	**274.3**	274.3	0.0
	TLBO	363.6	285.4	78.2
Dbl-Balanced	SA	2925.2	495.8	2429.4
	GA	2089.1	493.3	1595.8
	PSO	**1856.2**	516.0	1340.2
	TLBO	2742.4	492.9	2249.4
Dbl-Premium	SA	2803.8	502.8	2301.0
	GA	1919.9	469.9	1450.0
	PSO	**1254.5**	513.3	741.2
	TLBO	2453.3	472.8	1980.6

B. Complete Algorithm Comparison Including RL

Table IV extends the analysis to include the DQN agent across three double-fork case studies. The RL approach consistently achieves the fastest makespans (473–482 min) but incurs 183–199% higher charging costs than PSO due to the reward function weighting completion heavily ($R_{\text{complete}} = 100$) relative to charging cost penalties ($\lambda = 1$).

TABLE IV: All Algorithms on Double-Fork Scenarios

Scenario	Alg.	f	Mksp (min)	Cost (EGP)
Baseline (3)	SA	2885.0	497.1	2388.0
	GA	1805.8	475.5	1330.2
	PSO	**1483.6**	520.7	962.9
	TLBO	2386.7	486.1	1900.6
	RL	4697.9	**473.1**	4224.7
Congested (6)	SA	3495.1	551.5	2943.6
	GA	**2321.9**	487.2	1834.7
	PSO	3447.9	562.5	2885.4
	TLBO	3150.9	500.8	2650.1
	RL	5868.1	**468.2**	5399.9
High Dem. (7)	SA	4657.3	505.5	4151.8
	GA	**2851.3**	486.7	2364.6
	PSO	4073.0	502.9	3570.0
	TLBO	3644.5	495.7	3148.7
	RL	7563.9	**482.1**	7081.8

Fig. 2 visualizes the four-algorithm benchmark on a representative double-fork scenario, showing PSO's advantage in weighted objective and total cost alongside GA's competitive makespan. Fig. 3 provides a comprehensive breakdown across all three case studies including queue time, charging time, and energy delivered, revealing that SA consistently incurs the highest queue and charging times while GA maintains the lowest across congested scenarios.

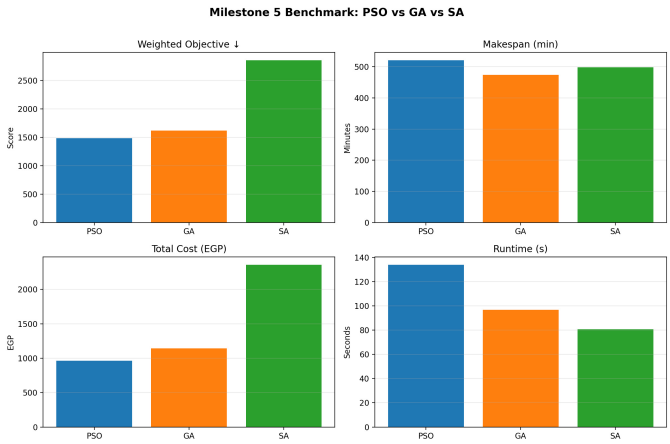

Fig. 2: Benchmark comparison of PSO, GA, and SA on the double-fork baseline. PSO achieves the lowest weighted objective and cost.

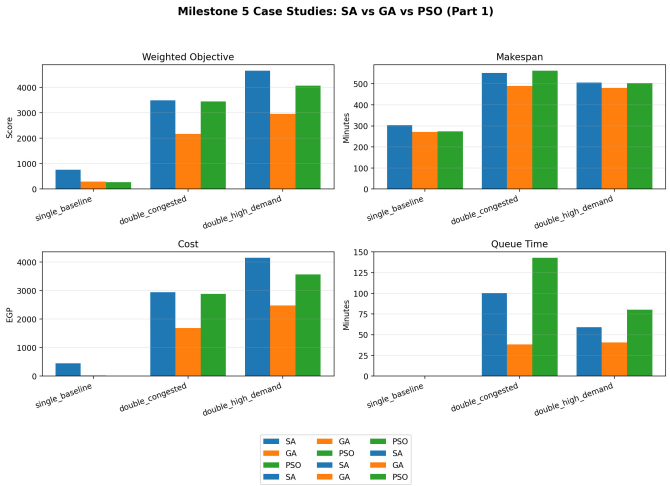

Fig. 3: Case study comparison (SA vs. GA vs. PSO) across single baseline, congested grid, and high-demand scenarios showing weighted objective, makespan, cost, and queue time.

C. PSO Convergence and Hyperparameter Sensitivity

Fig. 4 shows the PSO convergence trace on the double-fork scenario, exhibiting steady objective descent from 2500 to below 1500 before plateauing near iteration 150. This motivates the adaptive inertia schedule demonstrated in Fig. 1.

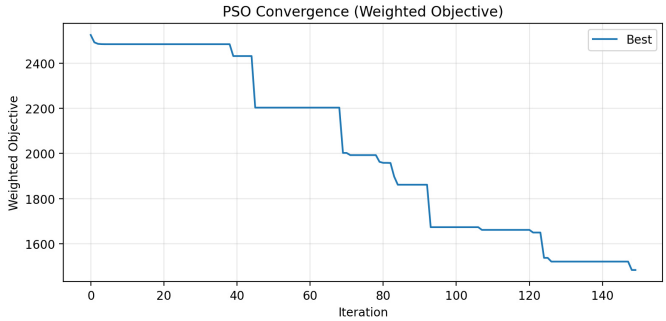

Fig. 4: PSO convergence on the double-fork scenario. The objective improves steadily before flattening near iteration 150.

A grid search over 27 hyperparameter configurations (Fig. 5) explores inertia $w \in \{0.4, 0.6, 0.8\}$, $c_1 = c_2 \in \{1.5, 1.7\}$, and swarm sizes $\{30, 45, 60\}$. The best configuration ($w = 0.6$, $c_1 = c_2 = 1.5$, swarm 45) achieves 521.97 min makespan with 525 EGP cost. A broader per-case sweep (243 trials) confirms that light inertia ($w = 0.3$) with symmetric cognitive/social terms dominates when iteration budgets increase.

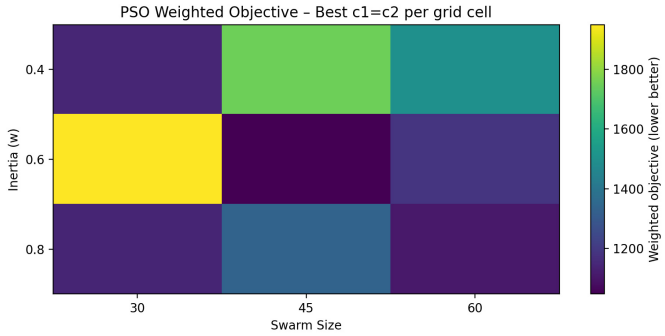

Fig. 5: PSO hyperparameter grid search heatmap showing weighted objective across inertia weights and swarm sizes. Darker cells indicate lower (better) objectives.

D. Final Five-Algorithm Comparison

Fig. 6 presents the comprehensive comparison of all five algorithms across the three double-fork case studies. Key findings include:

- **PSO dominates uncongested scenarios**, achieving 18–40% lower weighted objectives than the next-best algorithm by minimizing charging costs.
- **GA shows superior robustness under congestion**, maintaining consistent performance as fleet size increases from 3 to 7 EVs.
- **TLBO provides a parameter-free alternative**, consistently outperforming SA while requiring no tuning effort.
- **RL achieves the fastest makespans** (1–5% faster than best metaheuristic) but exhibits the highest total costs due to insufficient reward balancing.

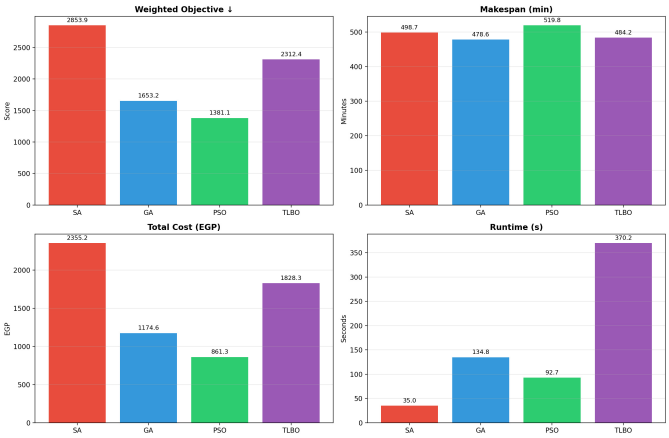

Fig. 6: Five-algorithm comparison across double-fork scenarios. GA maintains the lowest cost under congestion, while RL has the highest cost but lowest queue times.

E. Parameter Sensitivity Analysis

Table V summarizes the impact of parameter tuning. GA performance improves monotonically with population size (10% gain). PSO's balanced exploration-exploitation yields the best results, with pure exploitation degrading performance

by 57% due to premature convergence. TLBO benefits from larger populations (10% improvement), confirming that more learners enhance the teaching dynamics. Notably, extended RL training (1000 episodes) paradoxically worsens performance, suggesting overfitting.

TABLE V: Parameter Sensitivity: Weighted Objective by Configuration

Algorithm	Config	f	Cost (EGP)
GA	Small Pop	2132.4	1647.0
	Balanced	2020.0	1524.4
	Large Pop	**1895.4**	1403.9
PSO	Exploration	2156.5	1624.9
	Balanced	**1856.2**	1340.2
	Exploitation	2920.3	2418.5
TLBO	Small Pop	2845.3	2352.8
	Balanced	2675.7	2181.8
	Large Pop	**2567.7**	2066.8
RL	Quick (250)	5097.9	4659.4
	Balanced (500)	**4851.2**	4411.7
	Deep (1000)	5584.5	5150.0

VI. Discussion

The experimental results reveal a clear *no-free-lunch* pattern in EV fleet optimization: no single algorithm dominates across all scenarios. PSO's swarm-based exploration excels when charging decisions are the primary bottleneck (uncongested scenarios), but its performance degrades significantly under station congestion—from best to third-best—because queue penalties create a rugged fitness landscape that swarm dynamics struggle to navigate.

GA's population-based approach with route-diversifying crossover operators provides the mechanism to escape these congested local minima, making it the preferred choice for peak-demand operations. TLBO's performance, while not the best in any scenario, is remarkable given its complete absence of tunable hyperparameters—a significant practical advantage for fleet operators without optimization expertise.

The RL agent's behavior—achieving fastest makespans at exorbitant charging costs—highlights a fundamental challenge in reward engineering for multi-objective problems. The learned policy effectively optimizes for rapid network traversal but fails to develop nuanced charging strategies within the 500-episode training budget.

A promising future direction is *hybrid optimization*, using GA to quickly locate diverse high-quality candidate solutions, then refining each with SA's intensive local search. Additionally, combining GA mutation operators with PSO velocity updates may help overcome queue-induced congestion penalties.

VII. Conclusion

This paper presented a comprehensive comparative study of five optimization approaches for cooperative EV fleet routing and charging. The key algorithm selection guidelines for practitioners are:

1) **Uncongested operations:** PSO (18–40% improvement over alternatives).
2) **Congested/high-demand periods:** GA (robust to queue constraints).
3) **Limited tuning resources:** TLBO (parameter-free, competitive results).
4) **Time-critical applications:** RL (fastest makespan, requires reward engineering for cost control).

The adaptive inertia schedule for PSO improves convergence by 16% and should be adopted as a default strategy. Future work will investigate stochastic travel times, Pareto-optimal formulations, and hybrid GA–PSO strategies that combine population diversity with swarm refinement.

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